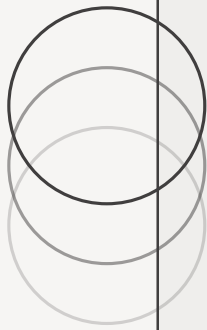


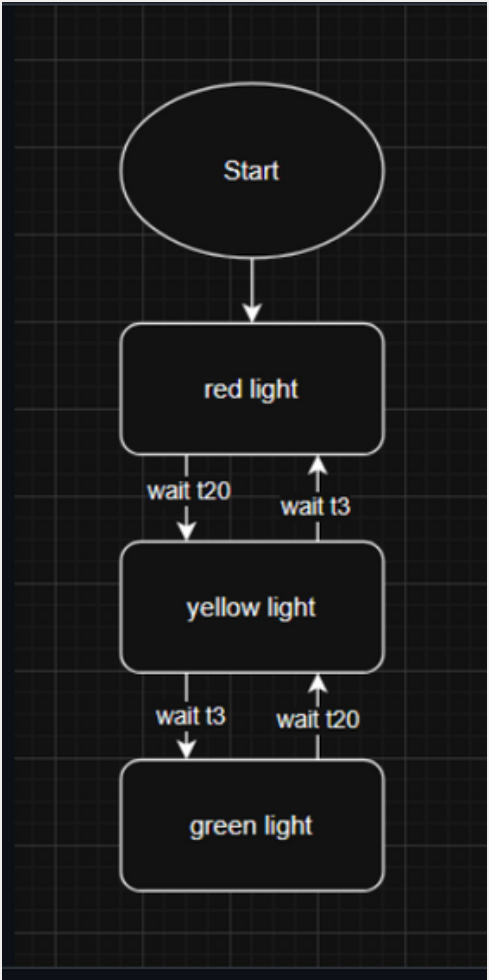
EMBEDDED SYSTEM

NGUYEN, LUKAS, NAHIN

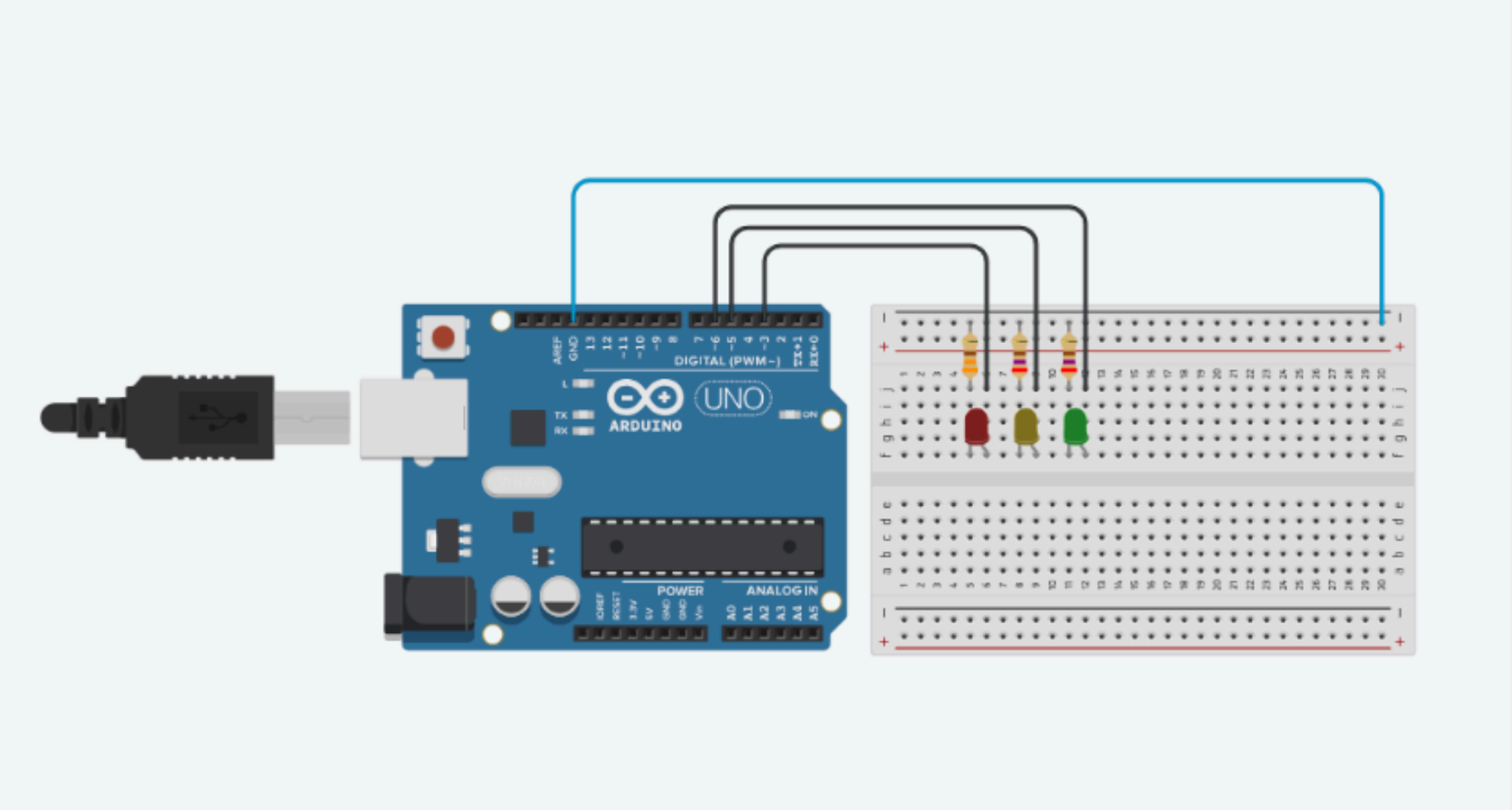
- EMBEDDED SYSTEMS:
 - TASK 1 TO TASK 5 OVERVIEW



DIAGRAM

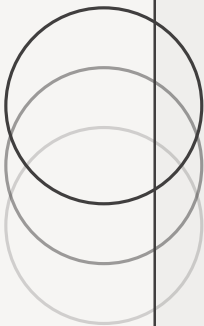


TINKERCAD SIMULATION



- EMBEDDED SYSTEMS:
-TASK 1

HOCHSCHULE HAMM-LIPPSTADT



CODE

```
// C++ code
//
unsigned int light = 0;
bool dir = 0;

void setup()
{
  pinMode(3, OUTPUT);
  pinMode(5, OUTPUT);
  pinMode(6, OUTPUT);
}

void loop()
{

  switch(light){
  case 1://red light
    analogWrite(5, 0);
    analogWrite(3, 192);
    delay(5000);
    dir = 1;
    light = 2;
    break;

  case 2://orange light
    analogWrite(3, 0);
    analogWrite(6, 0);
    analogWrite(5, 192);
    delay(2000);

    if(dir == 0){
      light = 1;
    }else{
      light = 3;
    }

    break;

  case 3://green light
    analogWrite(5, 0);
    analogWrite(6, 192);
    delay(5000);
    dir = 0;
    light = 2;
    break;

  default:
    light = 3;
    break;
  }
}
```

GITHUB

Embedded_LaboratoryPublic

Watch0Fork0Star0

embedded_task13 Branches0 Tags

Go to file

Add fileCode

About

This branch is 2 commits ahead of main.

Contribute

iamRustworthyAdd files via upload178b6e1 · 2 months ago3 Commits

Class_Diagram.jpg	Add files via upload	2 months ago
README.md	Initial commit	2 months ago
State_machine_diagram.png	Add files via upload	2 months ago
mikrocontroller.png	Add files via upload	2 months ago
task11.ino	Add files via upload	2 months ago

README

Embedded_Laboratory

Ben, Nahin, Lukas

Readme

Activity

0 stars

0 watching

0 forks

Report repository

Releases

No releases published

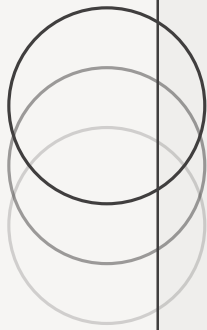
Create a new release

Packages

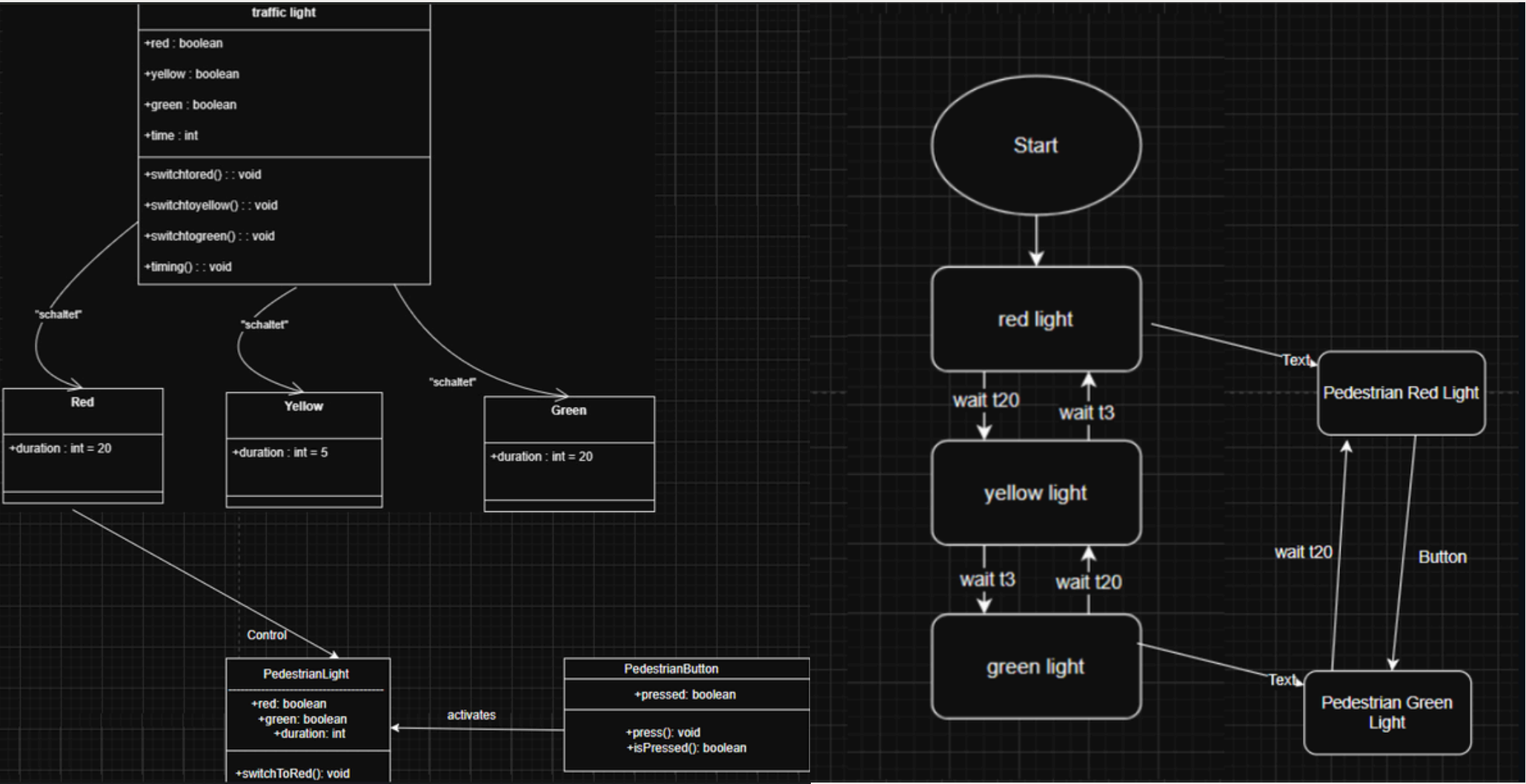
No packages published

Publish your first package

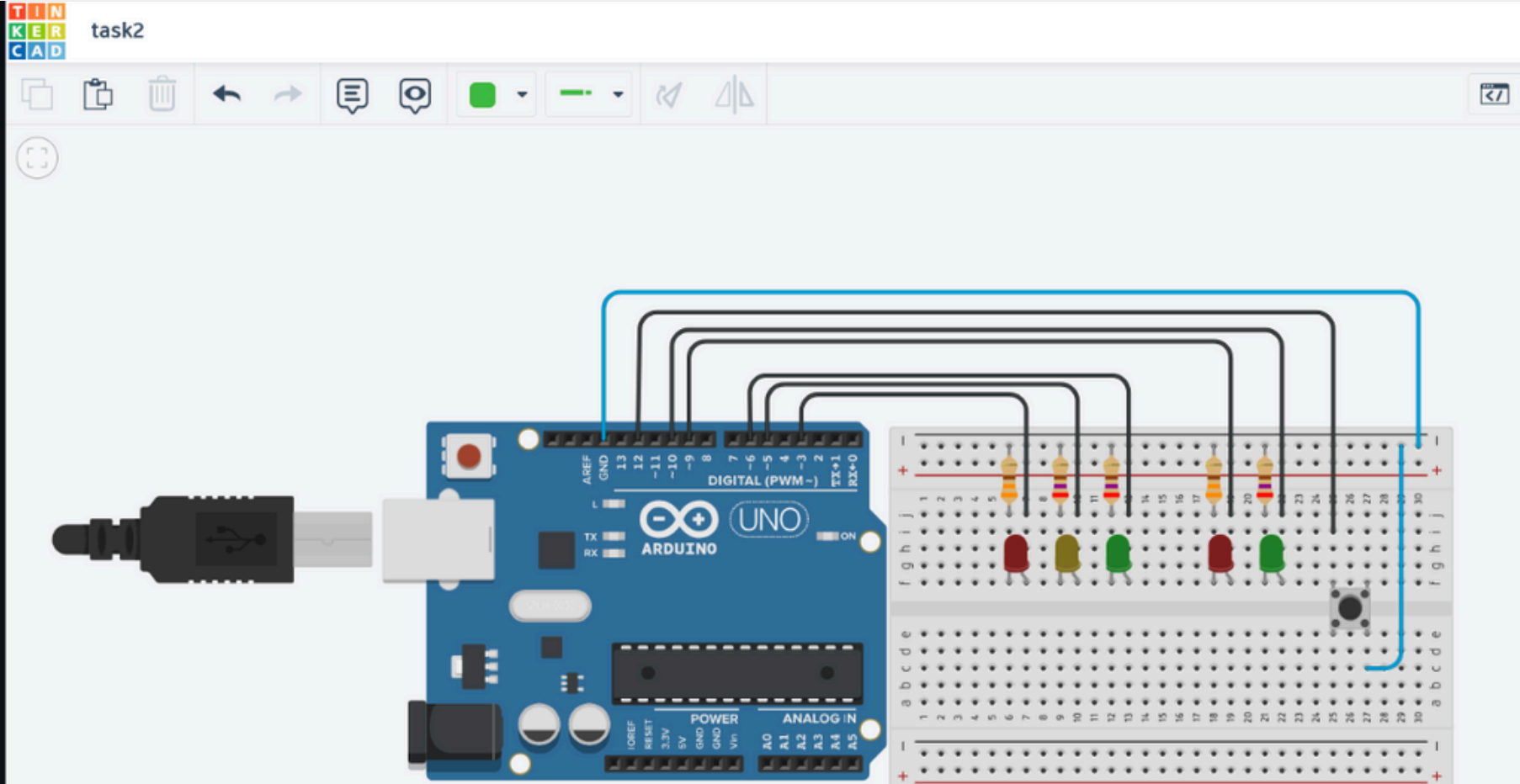
- EMBEDDED SYSTEMS:
-TASK 1



DIAGRAM

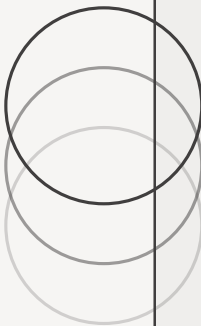


TINKERCAD
SIMULATION



- EMBEDDED SYSTEMS:
-TASK 2

HOCHSCHULE HAMM-LIPPSTADT



CODE

```
// C++ code
//
unsigned int light = 0;
bool dir = 0;

void setup()
{
  pinMode(3, OUTPUT);
  pinMode(5, OUTPUT);
  pinMode(6, OUTPUT);
  pinMode(9, OUTPUT);
  pinMode(10, OUTPUT);
  pinMode(12, INPUT_PULLUP); // Input with Pull up Resistor
}

void loop()
{
  switch(light){
    case 1: // red light
      analogWrite(5, 0);
      analogWrite(3, 192);
      delay(5000);
      dir = 1; // direction back
      light = 2; // go to traffic yellow
      break;

    case 2: // orange light
      analogWrite(3, 0);
      analogWrite(6, 0);
      analogWrite(5, 192);

      if(dir == 0){ // if direction coming from traffic red
        light = 5; // go to pedestrian green
      } else {
        light = 4; // go to pedestrian red
      }

      break;

    case 3: // green light
      analogWrite(5, 0);
      analogWrite(6, 192);
      delay(5000);
      dir = 0;
      light = 0; // go to button
      break;

    case 4: // ped red
      analogWrite(9, 192);
      analogWrite(10, 0);
      delay(2000);
      light = 3; // go to traffic green
      break;

    case 5: // ped green
      delay(2000);
      analogWrite(9, 0);
      analogWrite(10, 192);
      light = 1; // go to traffic red
      break;

    default: // is button pressed?
      while(bool button = digitalRead(12) == true){
        light = 2;
      }
      break;
  }
}
```

GITHUB

Embedded_LaboratoryPublic

Watch0Fork0Star0

embedded_task23 Branches0 Tags

Go to file

Add fileCode

About

This branch is 3 commits ahead of main.

Contribute

TienNguyenHSHLAdd files via uploade007b93 · 2 months ago4 Commits

Class_Diagram.pngAdd files via upload2 months ago

README.mdInitial commit2 months ago

State_machine_diagram.pngAdd files via upload2 months ago

mikrocontroller_task2.pngAdd files via upload2 months ago

task21.inoAdd files via upload2 months ago

README

Embedded_Laboratory

Ben, Nahin, Lukas

Ben, Nahin, Lukas

Readme

Activity

0 stars

0 watching

0 forks

Report repository

Releases

No releases published

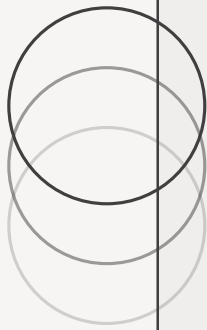
Create a new release

Packages

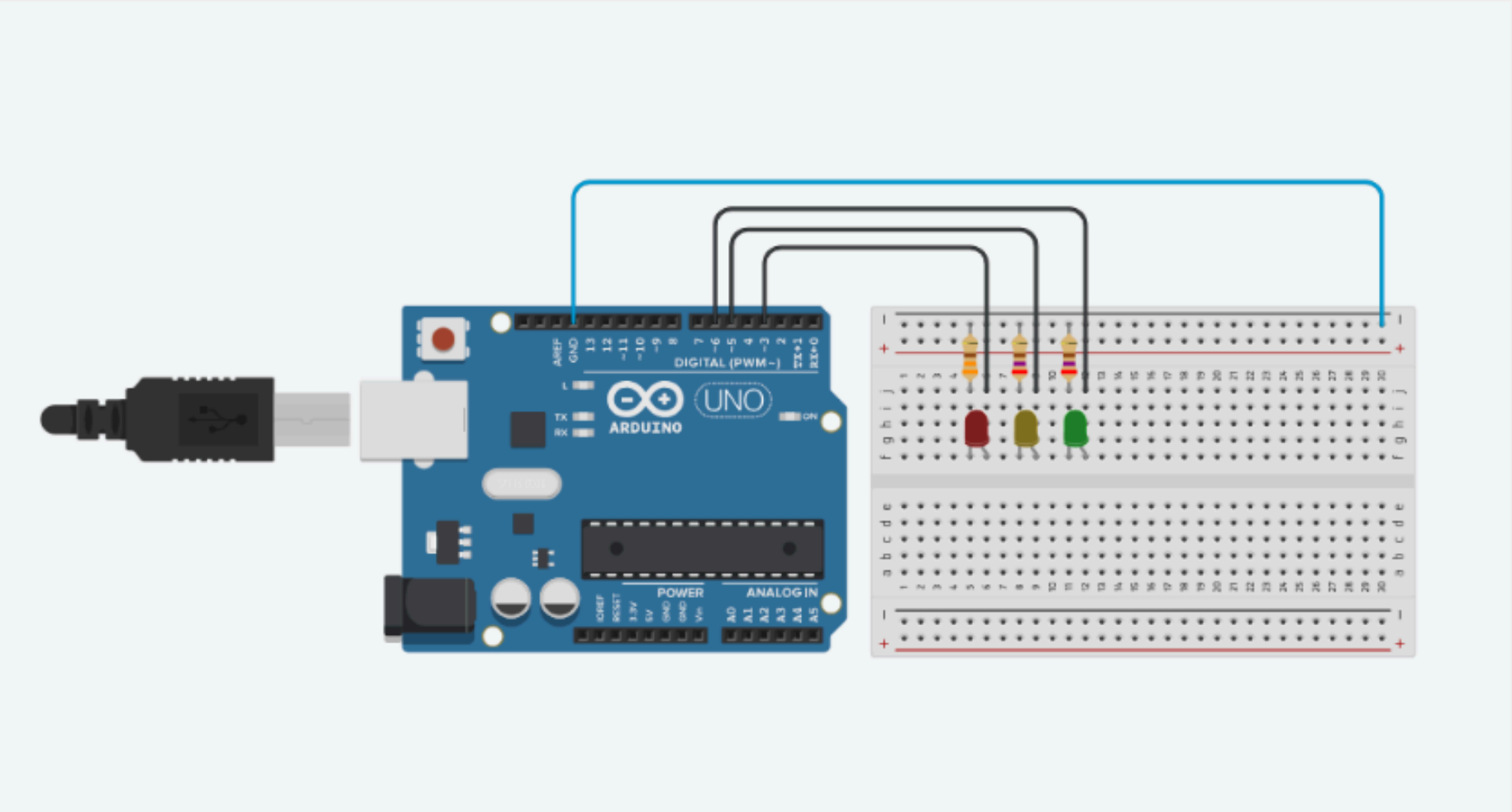
No packages published

Publish your first package

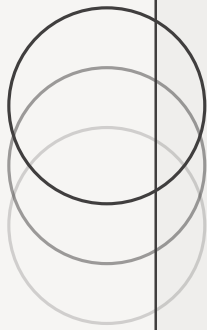
- EMBEDDED SYSTEMS:
-TASK 2



CIRCUIT



- EMBEDDED SYSTEMS:
-TASK 3



CODE

```
// C++ code
//
unsigned int light = 0;
bool dir = 0;

void setup()
{
  pinMode(3, OUTPUT);
  pinMode(5, OUTPUT);
  pinMode(6, OUTPUT);
}

void loop()
{

  switch(light){
  case 1://red light
    analogWrite(5, 0);
    analogWrite(3, 192);
    delay(5000);
    dir = 1;
    light = 2;
    break;

  case 2://orange light
    analogWrite(3, 0);
    analogWrite(6, 0);
    analogWrite(5, 192);
    delay(2000);

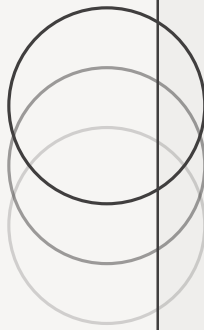
    if(dir == 0){
      light = 1;
    }else{
      light = 3;
    }

    break;

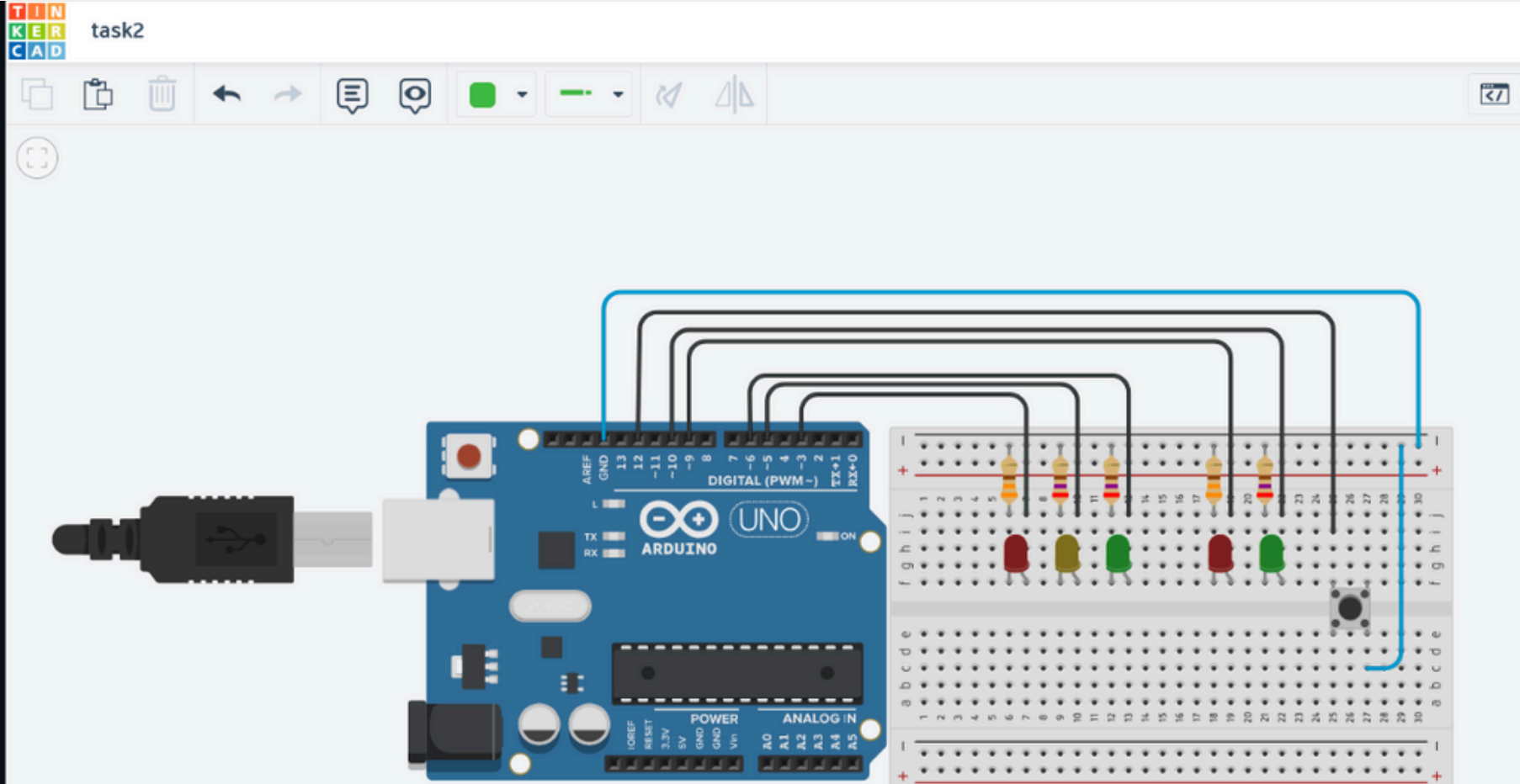
  case 3://green light
    analogWrite(5, 0);
    analogWrite(6, 192);
    delay(5000);
    dir = 0;
    light = 2;
    break;

  default:
    light = 3;
    break;
  }
}
```

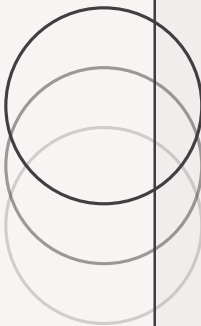
- EMBEDDED SYSTEMS:
-TASK 3



CIRCUIT



- EMBEDDED SYSTEMS:
-TASK 4



CODE

```
// C++ code
//
unsigned int light = 0;
bool dir = 0;

void setup()
{
  pinMode(3, OUTPUT);
  pinMode(5, OUTPUT);
  pinMode(6, OUTPUT);
  pinMode(9, OUTPUT);
  pinMode(10, OUTPUT);
  pinMode(12, INPUT_PULLUP); // Input with Pull up Resistor
}

void loop()
{
  switch(light){
    case 1: // red light
      analogWrite(5, 0);
      analogWrite(3, 192);
      delay(5000);
      dir = 1; // direction back
      light = 2; // go to traffic yellow
      break;

    case 2: // orange light
      analogWrite(3, 0);
      analogWrite(6, 0);
      analogWrite(5, 192);

    if(dir == 0){ // if direction coming from traffic red
      light = 5; // go to pedestrian green
    }else{
      light = 4; // go to pedestrian red
    }

    break;

    case 3: // green light
      analogWrite(5, 0);
      analogWrite(6, 192);
      delay(5000);
      dir = 0;
      light = 0; // go to button
      break;

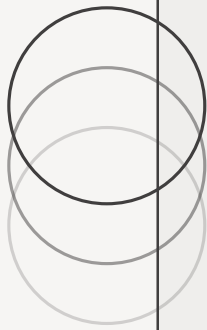
    case 4: // ped red
      analogWrite(9, 192);
      analogWrite(10, 0);
      delay(2000);
      light = 3; // go to traffic green
      break;

    case 5: // ped green
      delay(2000);
      analogWrite(9, 0);
      analogWrite(10, 192);
      light = 1; // go to traffic red
      break;

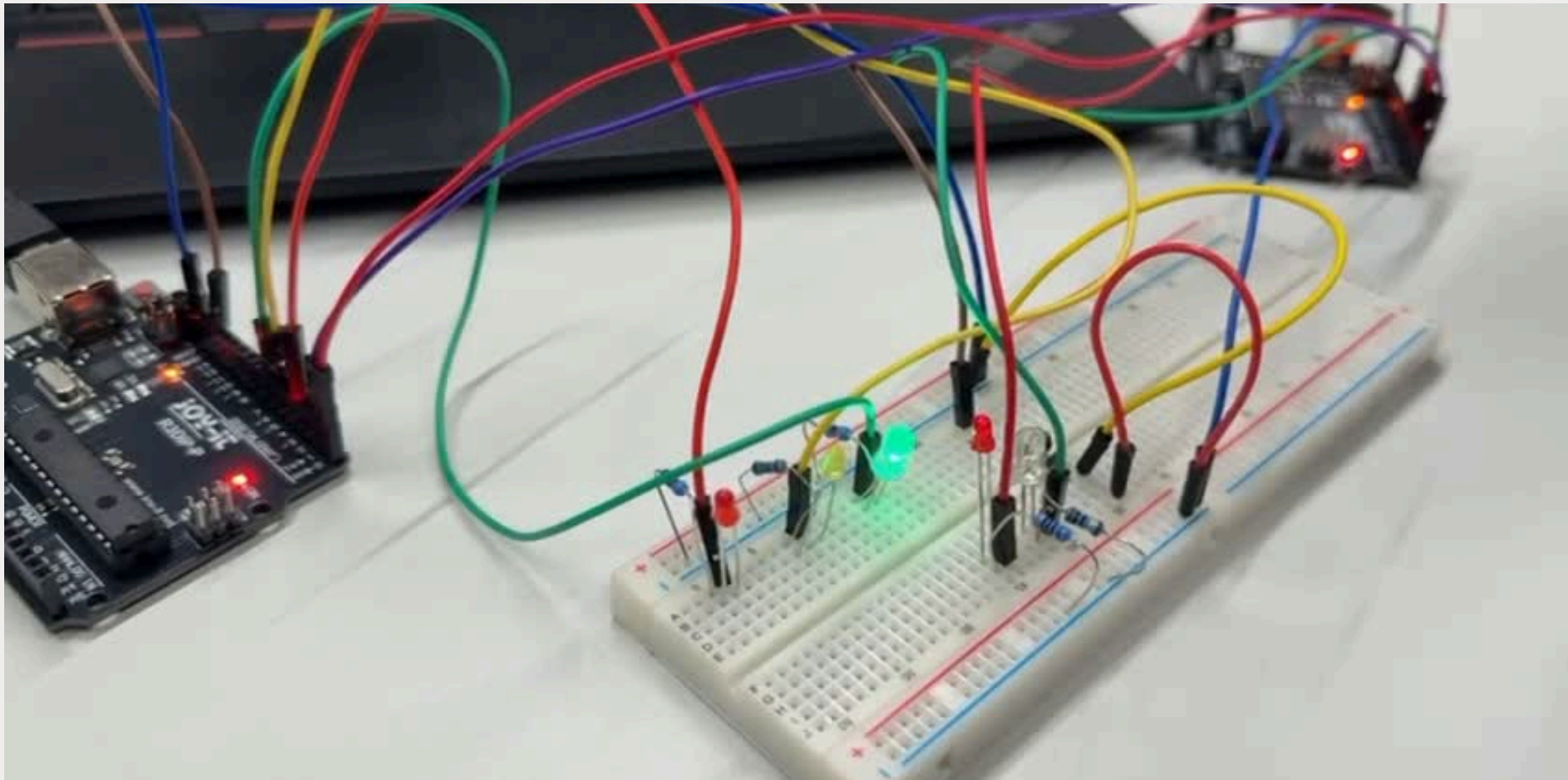
    default: // is button pressed?
      while(bool button = digitalRead(12) == true){
        light = 2;
      }
      break;

  }
}
```

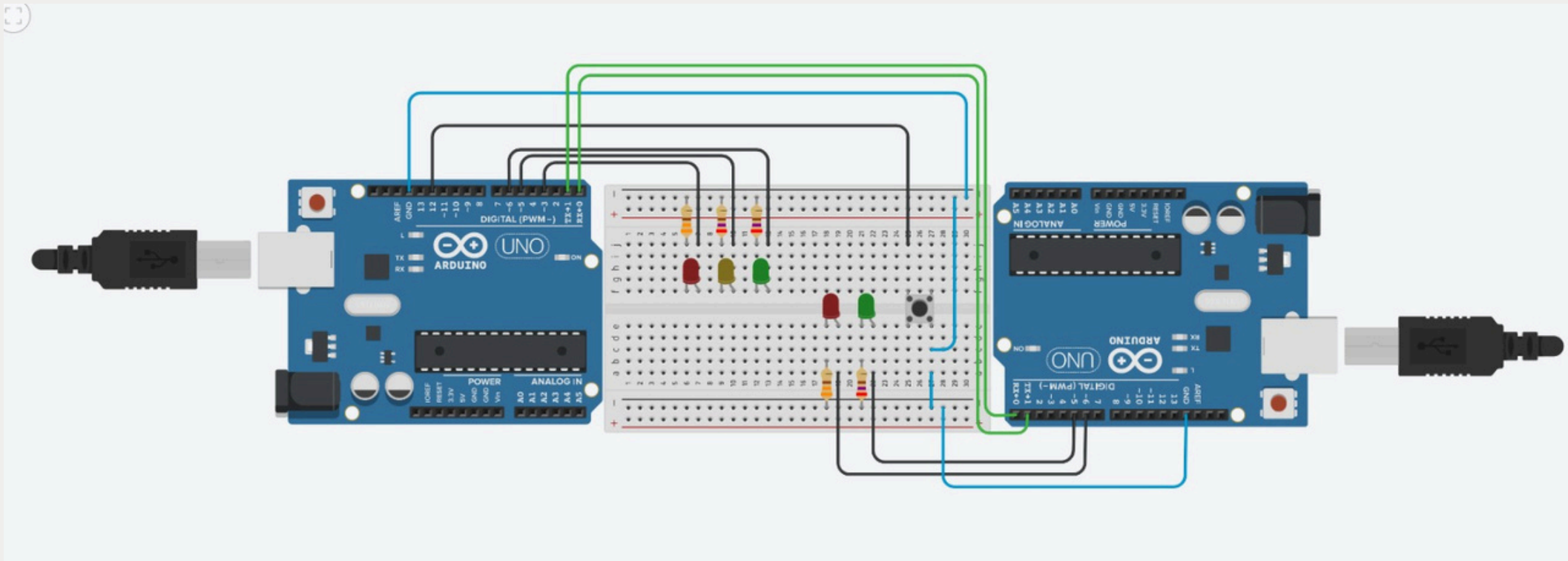
- EMBEDDED SYSTEMS:
-TASK 4



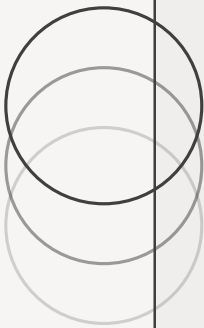
VIDEO



CIRCUIT



- EMBEDDED SYSTEMS:
-TASK 5



CODE

```
//MC1
UNSIGNED INT LIGHT = 0;

VOID SETUP() {
  PINMODE(3, OUTPUT); // RED
  PINMODE(5, OUTPUT); // ORANGE
  PINMODE(6, OUTPUT); // GREEN
  PINMODE(12, INPUT_PULLUP);
  SERIAL.BEGIN(9600);
}
VOID SETLIGHTS(INT R, INT O, INT G) {
  ANALOGWRITE(3, R);
  ANALOGWRITE(5, O);
  ANALOGWRITE(6, G);
}

VOID LOOP() {
  SWITCH(LIGHT) {

    CASE 0: // GREEN
      SETLIGHTS(0,0,192);
      IF (DIGITALREAD(12) == LOW) {
        LIGHT = 1; // GO TO ORANGE
      }
      BREAK;

    CASE 1: // ORANGE (ON THE WAY TO RED)
      SETLIGHTS(0,192,0);
      DELAY(2000);
      SERIAL.WRITE('R'); // REQUEST PEDESTRIAN CROSSING
      LIGHT = 2; // GO TO RED
      BREAK;

    CASE 2: // RED WAITING FOR PEDESTRIAN FINISH
      SETLIGHTS(192,0,0);
      IF (SERIAL.AVAILABLE() > 0) {
        CHAR C = SERIAL.READ();
        IF (C == 'D') { // PEDESTRIAN DONE
          LIGHT = 3; // GO TO ORANGE AGAIN
        } }
      BREAK;

    CASE 3: // ORANGE AFTER PEDESTRIAN CYCLE
      SETLIGHTS(0,192,0);
      DELAY(2000);
      LIGHT = 0; // GREEN AGAIN
      BREAK;

  }
}
```

```
//MC2
UNSIGNED INT LIGHT = 0;

VOID SETUP() {
  PINMODE(5, OUTPUT); // GREEN
  PINMODE(6, OUTPUT); // RED
  SERIAL.BEGIN(9600);
}

VOID SETPED(INT R, INT G) {
  ANALOGWRITE(6, R);
  ANALOGWRITE(5, G);
}

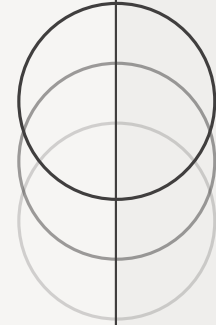
VOID LOOP() {

  SWITCH(LIGHT) {

    CASE 0: // PED RED
      SETPED(192, 0);
      IF (SERIAL.AVAILABLE() > 0) {
        CHAR C = SERIAL.READ();
        IF (C == 'R') { // REQUEST FROM MC1
          LIGHT = 1;
        }
      }
      BREAK;

    CASE 1: // PED GREEN
      SETPED(0, 192);
      DELAY(3000);
      SERIAL.WRITE('D'); // DONE
      LIGHT = 0;
      BREAK;

  }
}
```



THANKYOU

Please get in touch if you have any
questions or concerns about this report.

- EMBEDDED SYSTEMS:
-TASK 1 TO TASK 5 OVERVIEW